

Pseudo code of First Fit, Best Fit, Worst Fit

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#include <stdio.h>

void firstFit(int blockSize[], int blocks, int processSize[], int processes)
{
    int allocation[processes];

    for (int i = 0; i < processes; i++)
        allocation[i] = -1;

    for (int i = 0; i < processes; i++) {
        for (int j = 0; j < blocks; j++) {
            if (blockSize[j] >= processSize[i]) {
                allocation[i] = j;
                blockSize[j] -= processSize[i];
                break;
            }
        }
    }

    printf("\nFIRST FIT Allocation:\n");
    printf("Process No\tProcess Size\tBlock No\n");

    for (int i = 0; i < processes; i++) {
        printf("%d\t\t%d\t\t", i + 1, processSize[i]);

        if (allocation[i] != -1)
            printf("%d\n", allocation[i] + 1);
    }
}
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        else
            printf("Not Allocated\n");
    }
}

void bestFit(int blockSize[], int blocks, int processSize[], int processes)
{
    int allocation[processes];
    for (int i = 0; i < processes; i++)
        allocation[i] = -1;
    for (int i = 0; i < processes; i++) {
        int bestIdx = -1;
        for (int j = 0; j < blocks; j++) {
            if (blockSize[j] >= processSize[i]) {

                if (bestIdx == -1 || blockSize[j] < blockSize[bestIdx]) {
                    bestIdx = j;
                }
            }
        }
        if (bestIdx != -1) {
            allocation[i] = bestIdx;
            blockSize[bestIdx] -= processSize[i];
        }
    }
}

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}

printf("\nBEST FIT Allocation:\n");

printf("Process No\tProcess Size\tBlock No\n");

for (int i = 0; i < processes; i++) {

    printf("%d\t\t%d\t\t", i + 1, processSize[i]);

    if (allocation[i] != -1)

        printf("%d\n", allocation[i] + 1);

    else

        printf("Not Allocated\n");

}

}

void worstFit(int blockSize[], int blocks, int processSize[], int
processes) {

    int allocation[processes];

    for (int i = 0; i < processes; i++)

        allocation[i] = -1;

    for (int i = 0; i < processes; i++) {

        int worstIdx = -1;

        for (int j = 0; j < blocks; j++) {

            if (blockSize[j] >= processSize[i]) {

                if (worstIdx == -1 || blockSize[j] > blockSize[worstIdx]) {

                    worstIdx = j;

                }

            }

        }

    }

}

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    }
}

if (worstIdx != -1) {
    allocation[i] = worstIdx;
    blockSize[worstIdx] -= processSize[i];
}
}

printf("\nWORST FIT Allocation:\n");
printf("Process No\tProcess Size\tBlock No\n");
for (int i = 0; i < processes; i++) {
    printf("%d\t\t%d\t\t", i + 1, processSize[i]);
    if (allocation[i] != -1)
        printf("%d\n", allocation[i] + 1);
    else
        printf("Not Allocated\n");
}
}

int main() {
    int blocks, processes;

    printf("Enter number of memory blocks: ");
    scanf("%d", &blocks);

    int blockSize[blocks];

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printf("Enter sizes of memory blocks:\n");
for (int i = 0; i < blocks; i++) {
    scanf("%d", &blockSize[i]);
}

printf("Enter number of processes: ");
scanf("%d", &processes);

int processSize[processes];

printf("Enter sizes of processes:\n");
for (int i = 0; i < processes; i++) {
    scanf("%d", &processSize[i]);
}

int block1[blocks], block2[blocks], block3[blocks];
for (int i = 0; i < blocks; i++) {
    block1[i] = blockSize[i];
    block2[i] = blockSize[i];
    block3[i] = blockSize[i];
}

firstFit(block1, blocks, processSize, processes);
bestFit(block2, blocks, processSize, processes);
worstFit(block3, blocks, processSize, processes);

return 0;
```

}

```
C:\Users\Devi\Desktop\lab6.e: x + v
Enter sizes of memory blocks:
100 500 200 300 600
Enter number of processes: 4
Enter sizes of processes:
212 417 112 426

FIRST FIT Allocation:
Process No    Process Size    Block No
1             212           2
2             417           5
3             112           2
4             426          Not Allocated

BEST FIT Allocation:
Process No    Process Size    Block No
1             212           4
2             417           2
3             112           3
4             426           5

WORST FIT Allocation:
Process No    Process Size    Block No
1             212           5
2             417           2
3             112           5
4             426          Not Allocated

Process returned 0 (0x0)  execution time : 149.286 s
Press any key to continue.
```

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C:\Users\Devi\Desktop\lab6.e: x + v
Enter sizes of memory blocks:
400 700 200 300 600
Enter number of processes: 4
Enter sizes of processes:
212 512 312 526

FIRST FIT Allocation:
Process No    Process Size    Block No
1             212           1
2             512           2
3             312           5
4             526          Not Allocated

BEST FIT Allocation:
Process No    Process Size    Block No
1             212           4
2             512           5
3             312           1
4             526           2

WORST FIT Allocation:
Process No    Process Size    Block No
1             212           2
2             512           5
3             312           2
4             526          Not Allocated

Process returned 0 (0x0)  execution time : 73.742 s
Press any key to continue.
|
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